Q1. In each time frame there are 8 slots of equal length, since the master clock frequency is 8.333kHz, the sample rate is

$$F_s = \frac{f}{s} \rightarrow \frac{8.333\text{kHz}}{8} = 1.04k\text{Hz}$$

The LSB represents the frame synchronisation bit, leaving the other 7 bits free to enable a maximum 7-bit resolution which can be reduced to 4 bits, leaving the other 3 set to zero as they're not being used. Therefore regardless of what the coding scheme is set to, the time frame will always contain 8 slots meaning the sample rate is constant.

The Nyquist Sampling Criterion states that the sample rate must be greater than or equal to twice the highest frequency component in the signal.

$$F_s \geq 2f_N$$

Since the sample rate is fixed, the highest frequency the input analog signal can be is half of the sample rate.

$$f_N = \frac{F_s}{2} \rightarrow \frac{1.04\text{kHz}}{2} = 520.81\text{Hz}$$

Therefore, the maximum input analog signal bandwidth can be 520.81Hz, this is to prevent aliasing, which is a phenomenon where the encoder isn't capable of sampling fast enough for the decoder to accurately represent the input value.

Q2.

a) Since the sample rate frequency is 1.04kHz,
b) We can find the frame width by

$$T_{frame} = 8 \times \frac{1}{f} \rightarrow 8 \times \frac{1}{8.333\text{kHz}} = 960\mu\text{s}$$

c) We can find the width of a data bit by

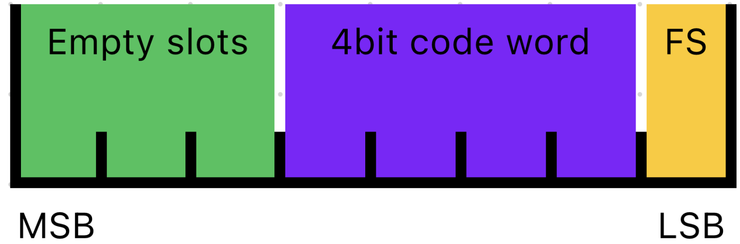
$$T_{bit} = \frac{1}{f} \rightarrow \frac{1}{8.333\text{kHz}} = 120\mu\text{s}$$

d) Since the coding scheme was set to 4 bits

$$T_{word} = T_{bit} \times 4 = 480\mu s$$

e) We found that there were 16 quantising levels, since the coding schedule was set to 4 bits
d) We did find the quantising levels were linearly spaced.

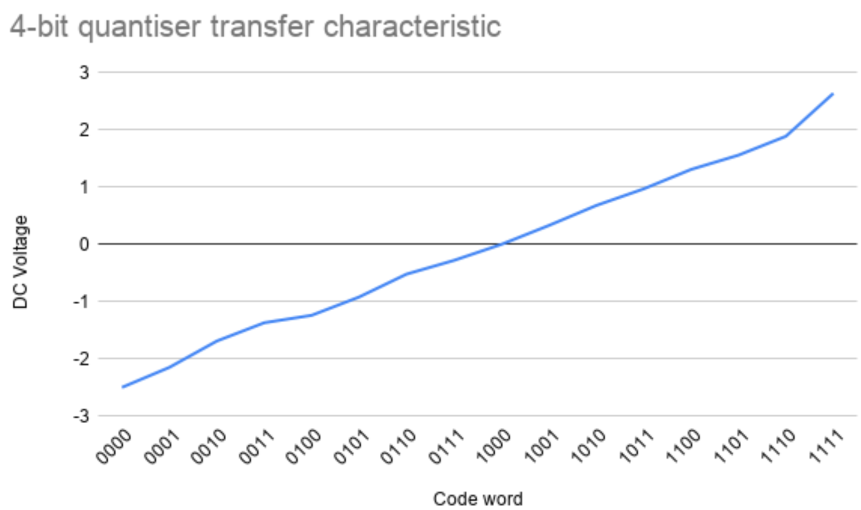
Q3. The data frame consists of 8 slots all equally spaced, a byte.



Q4. Yes it is possible to transmit each frame at a much slower rate than it was produced while still recovering the original message. This would be done by utilising a buffer between the encoder and the transmitter, this is to ensure no data-loss at the cost of real-time data being transmitted. The receiver would load new bits into a buffer until a full frame has been formed, before running into the decoder at the sampling rate. This may be an advantage when an application is forced to use a medium or a radio bandwidth that doesn't accommodate for the fast sampling rate such as a radio link or deep-space telemetry.

Q5. A DC message has a constant voltage, hence if it remains unchanged, the current code word will be the same as the next. A sine wave as an input is only unstable if the frequency of the message is not a sub-multiple of the frequency of the encoder clock. This is to ensure the encoder samples at the same phase point on the sine wave, without any synchronisation the sample will fall on different phase points.

Q6. To obtain this data, we slowly incremented the DC voltage input until we saw the code word change on the oscilloscope then we would record the voltage.



Q7. Since the clock rate is set to 8.333kHz and the frequency of the message is much lower in the hundreds, the passband can extend up to about 2kHz leaving anything above in the stop band that includes the spectral image of the clock frequency and any further harmonics of it. The passband needs to be flat too, to ensure all frequencies are treated the same and some bands aren't amplified more than others. A non-flat passband would give a false indication of performance as you'd be measuring the filters imperfections rather than the encoder/decoder itself.

Q8. Since the clock rate is set to 8.333kHz and the frequency of the message is much lower in the hundreds, the passband can extend up to about 2kHz leaving anything above in the stop band that includes the spectral image of the clock frequency and any further harmonics of it. The passband needs to be flat too, to ensure all frequencies are treated the same and some bands aren't amplified more than others. A non-flat passband would give a false indication of performance as you'd be measuring the filters imperfections rather than the encoder/decoder itself.

Q9. The sampling distortion comes from the hold duration taking too long which introduces distortion, especially with higher frequencies, similar to using a long shutter speed to capture a high speed object, it comes out distorted. To address this, you can reduce the hold duration to more accurately sample the message in the moment, another way is to use an equaliser which reverses the 'drooping' effect by amplifying the higher frequencies. With the quantising distortion using a 7-bit code word increases the quantising levels from 16 (4bit) to 128 which dramatically increases the resolution of the transmission which makes the PCM DATA output more accurate which reduces quantising distortion.

Q10. Typically with an increase of quantising levels the bit rate increases as it's the product of the bits per sample and samples transmitted per second. However with our experiment using the TIMS unit, the data frame has a constant 8 slots, so when switching between a 4bit and 7bit resolution the bit rate didn't change as we just utilise the 3 remaining empty slots.